ROYDAN CRUZ

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EDUCATION

Boston College, Chestnut Hill, MA

M.S in Data Science, *August 2026*

- Relevant courses: Multivariate Statistics, Applied Math for Data Science, Database Systems and Data Preparation, Applied AI and Machine learning, Deep Learning and AI

UMASS Lowell, **University of Massachusetts Lowell** Lowell, MA

B.A in Economics *December 2022*

Cumulative GPA: 3.92

- Honors / Awards: Summa Cum Laude, Omicron Delta Epsilon (ODE) International Honor Society for Economics member, MIT IDDEAS Scholar

RELEVANT COURSE PROJECTS

Causal Inference

Casual impact of financial literacy on financial well-being *Dec 2025*

GitHub Link: https://github.com/ApollocreedXI/Causal-Inferance

- Applied double machine learning to estimate the average treatment effect of financial literacy on financial well-being
- Double machine learning controlled the various within-covariate profiles to ensure only the treatment effect (i.e., whether someone was financially literate) impact on financial well-being was measured.
- No statistically significant causal effect was found, suggesting a more complicated relationship between financial well-being and financial literacy

Exploratory Data Analysis and Regularized Linear Regression

Home Value Prediction

Casual impact of financial literacy on financial well-being *Mar 2025*

GitHub Link: https://github.com/ApollocreedXI/Home-Value-Predictions/tree/main

- The purpose of this project was to develop a tool to analyze and detect undervalued real estate properties and gather insight into the features that are drivers of a property's value.
- This project revealed many drivers of a property's value, with ground living square footage and the quality of the home amongst the most influential.
- The lasso regression model explained a meaningful proportion of the variation in home sales price and had strong predictive accuracy. Thus, the tool would be useful for due diligence analysis in predicting a home's sales price after rehab repairs for ROI analysis.

WORK EXPERIENCE

Meran Tax Lawrence, MA

Accounting Assistant *Dec 2021 – Present*

- Manage the bookkeeping of 30+ small businesses from a variety of industries, including grocery stores, day care centers, restaurants, and more.
- Led a project to change our tedious manual bookkeeping process to an efficient semi-automated process. After implementation, we experienced a doubling in bookkeeping productivity. This led to meaningful labor cost savings for Meran Tax.
- Led a project to adopt OCR bank statement software for our bookkeeping process. This software adoption helped boost productivity by no longer resorting to manual entries of transactions when the file needed for semi-automation was not available.

Portsmouth Regional Hospital (HCA Health Care) Portsmouth, NH

Staff Accountant *Aug 2023 – Apr 2025*

- As a staff accountant, I was responsible for posting high-dollar impact journal entries for the New Hampshire market, reconciling month-end postings, managing our physician contract-based services, and working on other periodic tasks (e.g., Annual inventory audit, petty cash audits, cost reporting, and tax reporting).
- Created and worked with sophisticated Excel spreadsheets, using Power Query, to centralize data pulled from multiple sources. This streamlined the compliance review process, facilitating faster payment processing for our providers.
- Created a Power Query Excel spreadsheet to automate the data preprocessing of our corporate intercompany journal entries across the market. This Power Query automatically organized corporate postings by type and cleaned the transactional metadata. This led to clearer transaction descriptions and increased productivity in JE preparation.

TECHNICAL SKILLS

Programming and Data: Python, numpy, pandas, xarray, Database query optimization, Causal inference utilizing double machine learning

Data Visualization: Vega-Altair, seaborn, Streamlit

Modeling: sklearn, statsmodels, Multivariate statistical modeling, ML fairness evaluation, Time Series forecasting (ARIMA, GARCH LSTM Neural Nets), Traditional machine learning algorithmic modeling, Deep Neural Networks